

## WEEK-7

Task-1: To parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

CODE:

```
root@DESKTOP-QU0HVJI: ~/! × + v
GNU nano 7.2 input.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("input.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("Error opening input.txt");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading from input.txt:\n");

    while (fgets(buffer, sizeof(buffer), stdin))
        printf("%s", buffer);

    return 0;
}
```

OUTPUT:

```
root@DESKTOP-QU0HVJI:~/SKILL/skill-7# nano input.c
root@DESKTOP-QU0HVJI:~/SKILL/skill-7# gcc input.c -o input
root@DESKTOP-QU0HVJI:~/SKILL/skill-7# ./input
Error opening input.txt: No such file or directory
root@DESKTOP-QU0HVJI:~/SKILL/skill-7# |
```

Report: The input redirection program was successfully implemented in Ubuntu.

A file was opened using `open()` and connected to standard input using `dup2()`.

The program successfully read and displayed the contents of the input file.

Missing files were handled using an appropriate error message.

Task-2: To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1) {
        perror("Error opening output.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("Hello from output redirection!\n");
    printf("This data is stored in a file.\n");

    return 0;
}
```

Output:

```
root@DESKTOP-QU0HVJI:~$ mkdir output_project
  output_projectroot@DESKTOP-QU0HVJI:~$ cd output_project
root@DESKTOP-QU0HVJI:~/output_project$
root@DESKTOP-QU0HVJI:~/output_project$ nano output.c
root@DESKTOP-QU0HVJI:~/output_project$ gcc output.c -o output
root@DESKTOP-QU0HVJI:~/output_project$ ./output
root@DESKTOP-QU0HVJI:~/output_project$ output.txt
output.txt: command not found
root@DESKTOP-QU0HVJI:~/output_project$ cat output.txt
Hello from output redirection!
This data is stored in a file.
root@DESKTOP-QU0HVJI:~/output_project$ █
```

Report: The output redirection program was successfully implemented in Ubuntu. Standard output was redirected from the terminal to an output file using dup2(). The program created the file and stored the generated output inside it. The output data was verified using the cat command.

Task-3: To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

Code:

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("data.txt",
              O_WRONLY | O_CREAT | O_APPEND,
              0644);

    if (fd == -1) {
        perror("Error opening data.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("New line added.\n");

    return 0;
}

```

Output:

```

root@DESKTOP-QU0HVJI: ~/skill

root@DESKTOP-QU0HVJI:~$ mkdir skill
root@DESKTOP-QU0HVJI:~$ cd skill
root@DESKTOP-QU0HVJI:~/skill$ nano append.c
root@DESKTOP-QU0HVJI:~/skill$ gcc append.c -o append
root@DESKTOP-QU0HVJI:~/skill$ ./append
root@DESKTOP-QU0HVJI:~/skill$ cat data.txt
New line added.
root@DESKTOP-QU0HVJI:~/skill$

```

Report: The append redirection program was successfully implemented in Ubuntu. The file was opened using O\_APPEND so that new data was added at the end.

Existing file contents were preserved during multiple executions.  
The appended data was verified successfully using the cat command.

Task-4: TO parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

Code:

```
GNU nano 8.7.1 stderr.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("error.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1)
        return 1;

    dup2(fd, STDERR_FILENO);
    close(fd);

    FILE *file = fopen("missing.txt", "r");

    if (file == NULL) {
        perror("Error");
        return 1;
    }

    fclose(file);

    return 0;
}
```

Output:

```
root@DESKTOP-QU0HVJI:~/skill$ nano stderr.c
root@DESKTOP-QU0HVJI:~/skill$ gcc stderr.c -o stderr
root@DESKTOP-QU0HVJI:~/skill$ ./stderr
root@DESKTOP-QU0HVJI:~/skill$
root@DESKTOP-QU0HVJI:~/skill$
root@DESKTOP-QU0HVJI:~/skill$ cat error.txt
Error: No such file or directory
```

Report: The stderr redirection program was successfully implemented in Ubuntu.  
The standard error stream was redirected to a file using dup2().  
An intentional file-opening failure was generated to test error handling.  
The error message was successfully captured and verified inside error.txt.